

BANARAS HINDU UNIVERSITY

B.A./B.Sc. (Hons) — Semester VI Examination, 2022-23

Paper No. MTB-602: Linear Algebra

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Figures in right-hand margin indicate marks.*

Question 1. (Compulsory.)

- (a) Let V_1 and V_2 be subspaces of a vector space $V(F)$. Show that $V_1 + V_2$ is the subspace generated by $V_1 \cup V_2$. [3]
- (b) Let $T : V_1(F) \rightarrow V_2(F)$ be a linear transformation. Show that $\ker T = \{0\}$ if and only if T is one-one. [2]
- (c) Show that the vector space $F[x]$ of polynomials over F is not finite dimensional. [2]
- (d) Show that a matrix A and its transpose A^T have the same characteristic polynomial. [2]
- (e) Find the rank and nullity of the linear transformation $T : \mathbb{R}^3(\mathbb{R}) \rightarrow \mathbb{R}^3(\mathbb{R})$ defined by $T(x_1, x_2, x_3) = (x_1 + x_2 + x_3, x_2 + x_3, x_3)$. [3]
- (f) If x and y are vectors in an inner product space, show that:

$$\|x + y\|^2 + \|x - y\|^2 = 2(\|x\|^2 + \|y\|^2)$$

[2]

Question 2.

- (a) Define a vector space over a field. Can every Abelian group V be made into a vector space by the scalar multiplication $av = 0$ for all $a \in F$, $v \in V$? Justify your answer. [3]
- (b) If V_1 and V_2 are subspaces of $V(F)$, show that $V_1 \cup V_2$ is not a subspace in general. Further show that $V_1 \cup V_2$ is a subspace if and only if $V_1 \subseteq V_2$ or $V_2 \subseteq V_1$. [5]
- (c) Prove that any two bases of a finitely generated vector space $V(F)$ have the same number of elements. [6]

Question 3.

- (a) If W_1 and W_2 are two subspaces of a finite dimensional vector space $V(F)$, prove that:

$$\dim W_1 + \dim W_2 = \dim(W_1 + W_2) + \dim(W_1 \cap W_2)$$

[9]

- (b) (i) If $v_1, v_2, v_3 \in V(F)$ such that $v_1 + v_2 + v_3 = 0$, show that $\{v_1, v_2\}$ spans the same subspace as $\{v_2, v_3\}$. [3]
- (ii) Show by an example that linear dependence depends not only on the vector space but on the field as well. [2]
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Question 4.

- (a) State and prove the rank-nullity theorem. [7]
- (b) Define the dual space of a vector space $V(F)$. If $V(F)$ is an n -dimensional vector space with basis $\{e_1, e_2, \dots, e_n\}$, show that there exists a uniquely determined basis $\{e_1^*, e_2^*, \dots, e_n^*\}$ of its dual V^* such that $e_i^*(e_j) = \delta_{ij}$ for $j = 1, 2, \dots, n$, where $\delta_{ij} = 0$ if $i \neq j$ and $\delta_{ij} = 1$ if $i = j$. [7]
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Question 5.

- (a) If the matrix of a linear transformation T of $V_3(\mathbb{C})$ with respect to the basis $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ is

$$A = \begin{pmatrix} 0 & 1 & -1 \\ 0 & 1 & -1 \\ -1 & -1 & 0 \end{pmatrix}$$

find the matrix of T with respect to the basis $\{(1, 1, -1), (-1, 0, 1), (1, 2, 1)\}$. [8]

- (b) State the Cayley-Hamilton theorem. Verify it for the matrix:

$$A = \begin{pmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 3 & 1 \end{pmatrix}$$

[6]

Question 6.

- (a) Show that a square matrix A of order n is diagonalizable if and only if A has n linearly independent eigenvectors. [7]
- (b) Find the eigenvalues and eigenvectors of the matrix:

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & -4 & 2 \\ 0 & 0 & 7 \end{pmatrix}$$

[7]

Question 7.

- (a) If $\{u_1, u_2, \dots, u_n\}$ is any finite orthogonal set in an inner product space $V(F)$ and u is any vector in V , show that:

$$\sum_{k=1}^n |(u, u_k)|^2 \leq \|u\|^2$$

Furthermore, show that equality holds if and only if u is in the subspace generated by $\{u_1, u_2, \dots, u_n\}$. [8]

- (b) Apply the Gram-Schmidt process to vectors $y_1 = (1, 0, 1)$, $y_2 = (1, 0, -1)$, $y_3 = (0, 3, 4)$ to obtain an orthonormal basis for $\mathbb{R}^3(\mathbb{R})$ with standard inner product. [6]